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Integrated environmental and economic trade-offs in rice cultivation in emerging economies using a life cycle approach

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Abstract
Rice cultivation, a staple for over 70% of India's population and a major global export, significantly contributes to anthropogenic greenhouse gas emissions, alongside exacerbating water stress and facing climate-induced yield declines. This research uniquely integrates Life Cycle Assessment (LCA) with Life Cycle Cost Analysis (LCCA) to comprehensively evaluate environmental and economic trade-offs in conventional rice production across four diverse Indian agro-climatic regions–Assam, Haryana, Maharashtra, and Kerala. This comparative analysis identifies critical hotspots: increased logistical burdens from transport, elevated ecotoxicity from synthetic zinc sulphate, and the high cost of "safer" imported inputs. We also highlight a notable trade-off between decarbonization via reduced mechanization and increased labour intensity, which, while boosting rural employment, impacted production costs. These findings underscore the urgent need for evidence-based policies to achieve Sustainable Development goals on Climate Action, Zero Hunger, Clean Water and Sanitation, Decent Work and Economic Growth, and Life on Land. Effective interventions must focus on decentralized access to sustainable inputs, balanced subsidy reforms, stringent chemical regulation, and regionally tailored strategies to ensure sustainable and equitable rice production.

Keywords Life Cycle Assessment · Sustainable rice production · Emerging economies · Environmental hotspots · Agroecological trade-offs · Sustainable development goals

1 Introduction

At the turn of the last decade, global food systems are responsible for 13.6 Gt CO_2-eq (25%) of the 52.3 Gt of anthropogenic greenhouse gas emissions (GHGEs), of which 8 Gt CO_2-Eq comes from agricultural production [38, 55, 56]. In 2021, rice cultivation alone was the third highest contributor to food-related agricultural emissions (690 Mt CO_2-eq) after enteric fermentation (2.9 Gt CO_2-eq) and manure left on pasturelands (780 Mt CO_2-eq) [17]. Methane emissions from flooded rice farms contributed 30% of agricultural methane releases in addition to emissions from synthetic fertilisers, energy, manure, machinery, and fuel used at farm-level [32, 58]. Furthermore, the use of synthetic nitrogen fertilizer in rice production accounted for 11% of agricultural nitrous oxide (N_2O) emissions [23, 24]. Considering the total impact of

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these emissions, long-term projections indicate that rice cultivation could contribute up to 19% of global warming by the end of the twenty-first century [32, 34].

Abdo et al. [1] estimated the average carbon footprint of global rice production between 14 and 4854 kg CO_2-eq/tonne grain (averaging at 2430 kg CO_2-eq/tonne grain) attributing the wide range to differences in farm management, input, and system boundaries. Two-thirds of these emissions came from South and South-East Asian rice producing regions. Additionally, the water-intensive nature of rice cultivation exacerbated regional water stress, posing further sustainability challenges [64]. However, rice plays a crucial role in global food security especially for import-dependent nations in Africa, the Middle-East, and in Asia. India dominated global rice trade by exporting 33 million metric tons (MMt) in 2024–25 out of its total annual production of 145 million metric tons, which is far ahead of Thailand (11 MMt) and Vietnam (8.6 MMt) [17, 6, 69]. Domestically, rice remains a dietary staple for over 70% of India's population, with most of the harvest consumed within the country, highlighting its significance for policy initiatives such as the National Food Security Mission [71].

Rice production in India is both a contributor and a victim of climate change. Between 2015 and 2021, extreme weather events such as floods and droughts affected 33.9 and 35 million hectares of Indian cropland, respectively [5]. Future climate projections suggest that rice yields could decline by 3–10% under different emission scenarios by 2050, and could contribute to hunger among the most vulnerable [53, 66]. In particular, rainfed rice yields in India are projected to decrease by up to 20% by 2050 and by nearly 47% by 2080 [48]. A meta-analysis by Yu et al. [77] showed that elevated CO_2 in the atmosphere raised CH_4 emissions by 23% in rice fields in the short term, further promoting climate change.

Considering that the global population is expected to reach 10 billion by 2050 and India's population is projected to rise to approximately 1.7 billion, addressing the environmental footprint of this important crop is essential for ensuring long-term food security for over half the global population [36, 67]. Current research on rice cultivation focuses on solutions for improving yields and reducing Greenhouse Gas Emissions (GHGE), but this narrow perspective overlooks other interconnected environmental issues such as land degradation, water scarcity, biodiversity loss, and the socio-economic wellbeing of farming communities [9, 12, 19, 47, 76]. A comprehensive approach towards sustainable agriculture must incorporate these dimensions to avoid unintended trade-offs and promote resilient food systems.

Given the multifaceted challenges facing sustainable rice cultivation, a Life Cycle Assessment (LCA) offers a critical tool for understanding its comprehensive environmental footprint. The LCA methodology provides a robust framework for evaluating environmental impacts of a system across multiple categories, including climate change, water use, land use, eutrophication, and toxicity [27, 42]. By examining resource use and emissions throughout all stages of production, LCA enables identification of environmental hotspots and supports development of targeted mitigation measures. Unlike narrow metrics such as food miles, LCA offers a more comprehensive and regionally sensitive analysis [3]. This study also expands on traditional LCA by integrating a Life Cycle Cost Analysis (LCCA) to provide a more holistic understanding of both the environmental and economic dimensions of rice production.

Global LCA studies on rice production reveal geographic variations in environmental impacts, driven by differences in climate, soil properties, agricultural practices, and resource inputs [2, 20, 44]. In Asian rice-producing regions, methane emissions from flooded fields are the dominant source of environmental impact, whereas in other contexts, impacts related to energy use or nutrient runoff may be more significant [26, 36, 73]. Although several LCA studies have been conducted in rice-growing regions within India, systematic comparative analyses across multiple agro-ecological zones remain limited [37, 41]. This lack of comprehensive, regionally disaggregated data impedes the development of tailored mitigation strategies that account for local conditions.

This research introduces a first of its kind implementation of an integrated LCA and LCCA framework specifically applied to conventional rice cultivation. This was a standalone analysis under an attributional approach to provide a comprehensive understanding of both the environmental and life cycle cost dimensions across four rice-farms in distinct agro-climactic regions of India, without using a consequential approach. The analysis quantifies environmental impacts across key categories, identifies critical hotspots, compares environmental performance across individual farms, and evaluates the associated life cycle cost of rice production in these areas. The findings integrate the environmental and economic dimensions of rice cultivation in India, offering valuable insights for policymakers, researchers, and agricultural practitioners seeking to promote sustainable rice production.

2 Methodology

This research investigated the environmental impacts of conventional rice cultivation across four geographically distinct agro-climatic regions of India. For this study, a conventional rice farm was defined as an intensive rice production system characterized by its reliance on inorganic fertilizers and synthetic agrochemicals, being maintained under continuous, deep flooding (typically exceeding 10 cm) throughout the majority of the crop cycle, and may utilise mechanization for primary tillage and harvesting. The use of organic inputs such as farmyard manure was not excluded. Four rice farms, each ranging from three to seven acres in size, were selected from Assam, Haryana, Maharashtra, and Kerala. These sites were chosen to ensure maximum geographical and climatic diversity in the comparative analysis of environmental performance in India's primary rice-producing regions with an in-depth comparative assessment that captured the macro-level differences in impact profiles attributable to varied agro-ecological conditions and dominant farming practices. The farm located in Assam (F1) was in a humid subtropical climate characterized by high annual rainfall and a mean annual temperature of approximately 24 °C. In contrast, the farm in Kerala (F4) is situated in a region with high humidity, substantial rainfall, and a mean annual temperature of 27 °C. The farms in Haryana (F2) and Maharashtra (F3) are located in semi-arid and tropical climates, respectively, with mean annual temperatures of 25 °C and 26 °C. Haryana experiences hot summers and cold winters, while Maharashtra employs a combination of dam and rain-fed irrigation under moderate rainfall conditions. This regional selection facilitated a comparative assessment of rice cultivation's environmental performance under varied climatic, hydrological, and agricultural conditions (Fig. 1).

The research adhered to the Life Cycle Assessment (LCA) methodology standardized under ISO 14044:2006, encompassing four key phases: goal and scope definition, inventory analysis, impact assessment, and interpretation [33]. These phases were adapted to reflect the specific requirements of agricultural production systems, with a particular focus on rice cultivation. The environmental impacts were classified into three impact categories—resource utilization, human health effects, and ecological consequences—to provide a comprehensive and multi-dimensional assessment.

The goal of this LCA was to quantify the environmental impacts of conventional rice production from cradle to farm gate for the Kharif season across the four selected regions. This system boundary was selected based on the rationale that the majority of environmental burdens associated with rice production occur during agricultural operations prior to post-harvest processing [31]. This cradle-to-gate approach included inputs during seed preparation, land preparation, fertilizer and pesticide use, irrigation, machinery operation, and manual labour to the point of paddy threshing at the farm gate (Fig. 2). Subsequent stages such as milling, storage, and distribution were excluded, as these processes tend to be mechanized and exhibit limited regional variation [30, 55]. This approach allowed a focused comparison of on-farm environmental performance across the four research regions, where variations in farming practices and resource use are most pronounced [26, 57, 73].

The functional unit (FU) for this comparative Life Cycle Assessment is defined as the production of one tonne (1000 kg) of threshed paddy rice (*Oryza sativa L.*) with a standard moisture content of 14%. The spatial and temporal scope covers

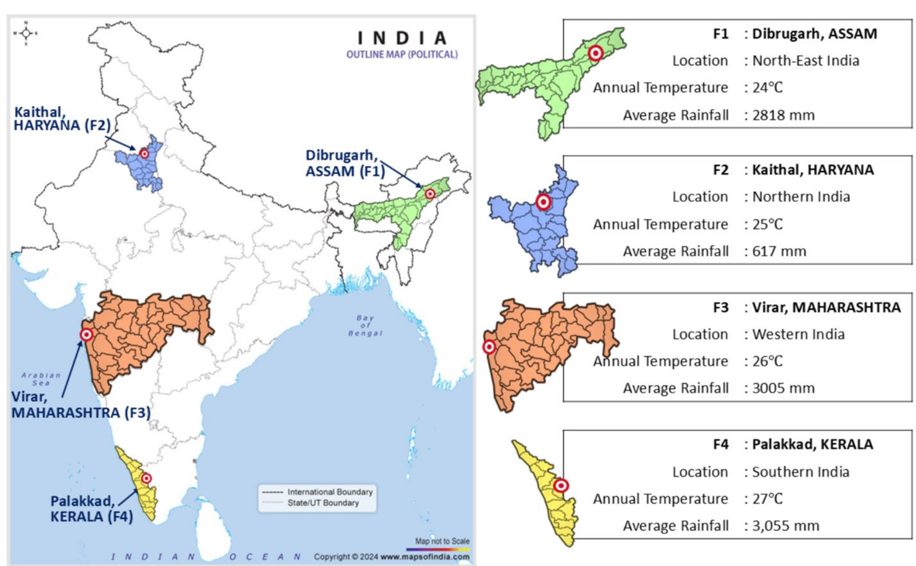

Fig. 1 Political map of India showing the location of the selected farms, F1, F2, F3, and F4 (Source: www.mapsofindia.com. Representational map only. Not to scale).

Fig. 2 System Boundary and farm inputs for the functional unit (1000 kg paddy) for each of the four regions–Assam (F1), Haryana (F2), Maharashtra (F3), and Kerala (F4)

LIFE CYCLE STAGE	F1 Inputs	F2 Inputs	F3 Inputs	F4 Inputs	OUTPUT
SEED TREATMENT	Masuri Rice Seed, Jaggery, Cow Urine, Urea, UPL Saaf, Water	Pre-treated Seed, Shared Seed, Water	Sundar, Ratna Seed, Cowdung, Water	Ponni Rice Seed, Jaggery, Water	1000 kg paddy grains
SOIL PREPARATION	Diesel for Tractor, Cow Dung	Diesel for Tractor	Diesel for Power Tiller, Grease	Diesel for Tractor, Fuel for Rotavator Attachment	
SOWING AND TRANSPLANTATION	Human Labour	Human Labour, Pretilachlor (50%), Chlorpyrifos (50%), Isoprothiolane	Human Labour	Human Labour	
WATER MANAGEMENT	Water, Petrol for Pump	Water, Diesel for Pump	Water from Dam	Water, Electricity for Pump	
SOIL FERTILITY	NPP (15-46-0), NPK (20:20:20), Water, Muriate of Potash, Urea (46%)	Cowdung, Water, Urea (46%)	NPK (15:15:15), Water, Scavenged Cowdung, Urea (46%)	DAP (18% N, 40% K), NPKS (20:20:0:13), Urea (46%), Zinc sulphate monohydrate (33%)	
PLANT PROTECTION	Acetamiprid, Tobacco leaves, onion leaves	Thiamethoxam (30%), Chlorantraniliprole (18%), Bifenthrin (10%), Fipronil (0.6%), Carbendazim (12%), Mancozeb (63%)	No specific inputs.	EC Lambda-cyhalothrin (5%), Water	
HARVESTING	Human Labour	Human Labour	Human Labour	Human Labour, Use of Combine Harvester	
THRESHING	Diesel for Thresher	Diesel for Thresher	Manual Thresher	Diesel for Combine Harvester	

the Kharif cultivation season in 2024, across four distinct regions in India. This mass-based FU was chosen for consistent comparison, reflecting the environmental burden under single-season conditions, which integrates the dynamics of seasonal rainfall and supplemental irrigation without multi-year linearization.

The processes within ecoinvent database incorporated energy consumption for on-farm activities. Initial flood irrigation volume was quantified by multiplying the measured water depth by the cultivated area (Volume = land area × height or depth). Subsequent supplemental irrigation was performed when the water level subsided by about 50%, with farmers reporting an application frequency of approximately thrice weekly during the peak vegetative stage, followed by a controlled reduction during the reproductive and maturation phases of rice development. The total water input was determined by adding the initial volume and the calculated volumes of all subsequent irrigations.

Primary data were collected through field surveys and structured interviews with farmers. Data collection covered a range of farm characteristics, including land area, cultivation practices, input use, energy consumption, irrigation methods, and yields. Specific data points included quantities of seeds, fertilizers, pesticides, and herbicides used; machinery utilisation; irrigation; labour inputs for various farming activities; and yields of threshed paddy. Diesel and petrol use were converted into energy units (megajoules) for standardized reporting. Rainfall contributions were excluded from irrigation calculations to isolate artificial water use.

Environmental impact assessment was conducted using OpenLCA version 2.4, in conjunction with the ecoinvent database version 3.11 [16, 22]. An attributional approach using the Unit Process data was employed based on the Allocation at the Point of Substitution (APOS) model from the ecoinvent 3.11 database which determined how environmental burdens may be allocated to co-products through system expansion and substitution, and influenced the LCI results. Therefore, any negative values observed in the LCA results represents environmental credits generated upstream within the ecoinvent background data flows. The ReCiPe 2016 Midpoint (Hierarchist) model was selected due to its use of the Hierarchist (H) perspective, which provides a balanced approach to uncertainty and time horizon, translating detailed data from life cycle into quantifiable environmental damage scores and generating midpoint indicators that directly link specific agricultural emissions to three endpoint categories, namely, human health, ecosystem quality, and resource scarcity [20, 26, 28, 36]. The analysis covered a range of midpoint impact categories, including global warming potential, acidification, eutrophication, ecotoxicity, ozone depletion, carcinogenic and non-carcinogenic toxicity, ionizing radiation, particulate matter formation, photochemical oxidant formation, non-renewable energy use, land use, mineral and metal depletion, and water use. The assessment considered impact categories–human health, ecosystem quality, and resource depletion.

A key constraint in this LCA was the limited geographical representativeness of the background life cycle inventory (LCI) data used. While our foreground data is highly site-specific, the background data used for agricultural inputs (the burdens from cradle-to-gate production) relies heavily on global or generalized regional averages due to the scarcity of publicly available, verified LCI data for Indian manufacturing facilities. Specifically, flows for high-impact inputs such as complex NPK fertilizers (e.g., Diammonium Phosphate, Potassium Chloride) and active pesticide ingredients were sourced predominantly from Global (GLO) or Rest-of-World (RoW) datasets within the ecoinvent 3.11 database. This decision was necessary to maintain a complete system boundary and avoid the material exclusion of high-burden inputs, recognizing that the chemical supply chain for Indian agriculture is largely integrated into the global market.

Considering the complexity of LCA modelling and databases, an evaluation of the sensitivity of the LCA model to the inputs was performed using uncertainty analysis. This was implemented using a Monte Carlo Simulation (MCS), employing a robust number of 50,000 iterations at 95% confidence to ensure the reliability and statistical stability of the results. This extensive sampling allowed for the quantitative propagation of input data uncertainty [11].

In addition to the LCA, a life cycle costing analysis (LCCA) was conducted using the same functional unit (one tonne of rice produced) and system boundary (cradle-to-farm gate), and inventory data following the principles of Environmental Life Cycle Costing (E-LCC) as defined by Swarr et al. [63]. The LCCA quantifies the total monetary costs over the assumed lifespan of the production system, including expenditures on seeds, fertilizers, pesticides, machinery fuel, electricity, equipment rental, and labour. Input costs were documented in Indian Rupees (INR) based on actual expenditures reported by farmers during field surveys (Table 2). All costs are reported as cost to the farmer. The analysis classified costs into five categories:

i. Natural inputs: Cost of cow dung, tobacco leaves, jaggery, water, and rice seeds
ii. Chemical inputs: Cost of fertilizers, pesticides, insecticides, and fungicides, (including subsidized products)
iii. Labour costs: Cost of human labour for various field operations
iv. Fuel/ Energy cost: Costs for diesel and petrol, and electricity
v. Rental costs for machinery: Costs for hiring tractors, harvesters, and threshers.

The total LCC per functional unit was calculated as the sum of these direct costs. Cost of water was negligible, but costs of fuel for the water pumps was added in the energy costs. Due to the focus on operational expenses and data constraints related to legacy infrastructure, capital costs for assets exceeding one crop cycle (e.g., land purchase, permanent borewells, pumps) were excluded from the analysis. The LCCA is therefore a measure of variable operating costs required to produce one tonne of rice, rather than the full cradle-to-grave economic life cycle of the entire farm. Costs related to land area (e.g., purchase price or opportunity cost of capital) were excluded from this LCCA. This decision was made because the study focuses on the on-farm operational efficiency and monetary outflows directly controllable by the farmer for the production system, rather than investment strategy. All costs are reported as nominal costs (or current prices), representing actual expenditures in the year of the survey. Since the analysis solely covered costs incurred within a single, defined cropping season and machinery usage was accounted for as rental charges (thereby excluding the need for capital asset amortization), the use of a discount rate to account for the time value of money was not deemed relevant for this study's scope. The aggregated financial outlay provided a comprehensive assessment of the economic performance per tonne of rice produced in each region.

3 Results

Among the non-toxic midpoint indicators, terrestrial acidification potential exhibited substantial variation across the studied farms. The highest acidification impact was observed in Kerala (F4), with a value of 1.895 kg SO_2-equivalent, which was a little over twice the impact recorded in Assam (F1) and Maharashtra (F3). Haryana (F2) displayed intermediate acidification potentials (Fig. 3). The primary sources of acidification were the use of diesel-powered water pumps, manure from cattle reared for milk production, heat produced at industrial furnaces, and the production of sulfuric acid in intermediate processes during agrochemical production (Supplementary Table S2). The single-largest contributors to these impacts include processes in regions beyond India, reflecting the nature of global supply chains and background systems that support the availability of inputs used in Indian farms. While many agricultural inputs and machinery are produced in India, some require inputs such as crude oil, metals, or chemical components that may be sourced from other countries. As inputs that were purchased were coded as being sourced from the market system, the market datasets expressed the results as per the blend of domestic production and imports. However, this regional specificity may be limited, as the ecoinvent database does not have country-specific data for every process specific to the country. In such cases, proxies such as GLO (Global average), RoW (Rest of World) appear in the results.

Climate change potential, expressed as global warming potential (GWP), also revealed significant disparities. Based on an estimated 0.5-acre land use for producing one tonne paddy (as reported by the farmer), F4's (GWP 408.116 kg

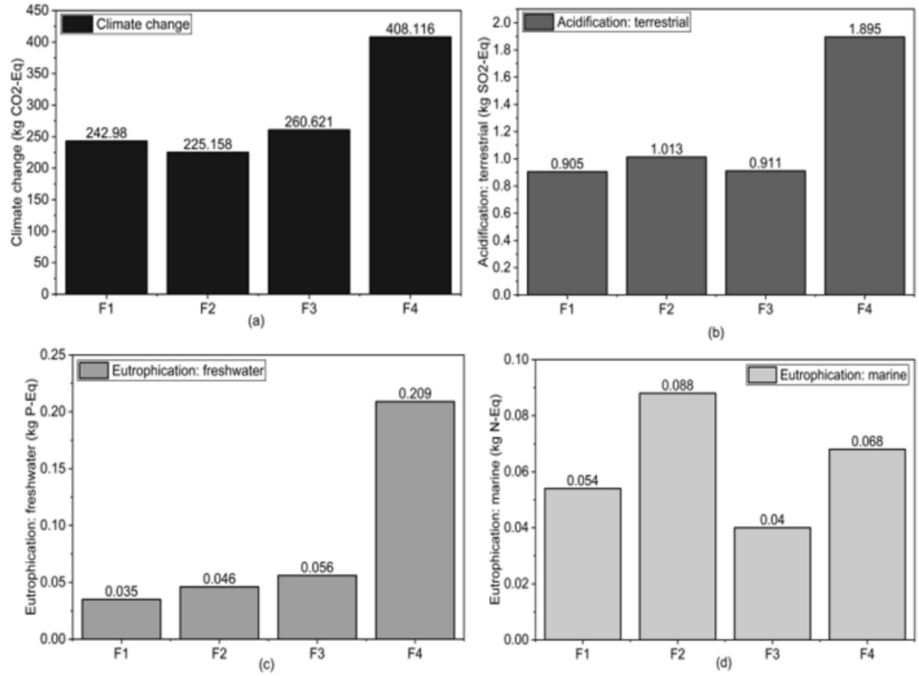

Fig. 3 Mid-point characterization results: **a** Climate Change or Global Warming Potential, **b** Acidification, **c** Freshwater Eutrophication, **d** Marine Eutrophication

CO_2-eq) area-based GWP (2.017 tCO_2-eq ha^{-1}) is approximately 22% lower than the average area-based GWP reported by Dash et al. [14] across their five contrasting rice production systems (~ 2.6 tCO_2-eq ha-1). Key contributors to GWP included diesel powered water pump operation, and the production of ammonia, diesel and natural gas, which may be involved in intermediate agrochemical production processes. These findings represent the input supply chain burden and highlighted the considerable influence of both farm practices and input sourcing on climate-related impacts.

Marine eutrophication was highest in F2, driven by silage production likely for animal feed, which reflects the co-products or multifunctionality in LCA. The runoff from the production of rice-seed was the highest contributor to marine eutrophication in the other three farms. F4 exhibited the highest freshwater eutrophication potential, primarily driven by processes related to the treatment of spoil from lignite mining and fertilizer production.

Energy resource consumption displayed considerable variability among farms. F4 had the highest fossil fuel use, where hard coal mine operations were the single largest contributor likely due to the use of electric water pumps and the electricity mix that depended heavily on coal-powered thermal plants (Fig. 4). In contrast, F1 and F2 demonstrated lower fossil fuel use, primarily related to fuel used in agricultural machinery. Water use followed a similar trend, with F4 consuming the most water, not only due to extensive pumped irrigation and water-intensive field management but also due to upstream processes involved in the production of rice-seed and other agricultural inputs, whereas F1 achieved significant water savings by combining rain-fed and partial irrigation strategies.

F4 also was attributed with the highest use of material resources in terms of metals and minerals (8.711 kg Cu-Eq) over twice more than F1 and F2, mainly from phosphate production.

Ecotoxicity impacts were also pronounced, and F4 consistently recorded the highest ecotoxicity potentials, with freshwater and marine ecotoxicity values markedly exceeding those of the other farms, while F2 represented the lowest in all three categories of ecotoxicity (Fig. 5). The largest contributors were upstream processes related to fertilizer and pesticide production, including sulphide tailings treatment, treatment of bottom ash, and the treatment of waste from metal mining and refining. Notably, F2 showed negative freshwater ecotoxicity values, indicating an environmental benefit or burden-shifting within the system's life cycle.

Terrestrial ecotoxicity impacts were especially severe in F4, primarily due to the use of zinc sulphate monohydrate, with substantial contributions arising from activities linked to zinc fertilizer production. Additionally, non-exhaust emissions from agricultural machinery, particularly brake and tire wear, contributed significantly to terrestrial ecotoxicity, particularly in mechanized farms where equipment usage was extensive. Although F3 used minimal mechanisation, it still used a 'power tiller' and a manual thresher that required regular greasing.

Human toxicity impacts, particularly carcinogenic impacts, were predominantly driven by treatment of industrial by-products such as electric arc furnace slag and non-carcinogenic impacts were driven by the treatment of sulfidic

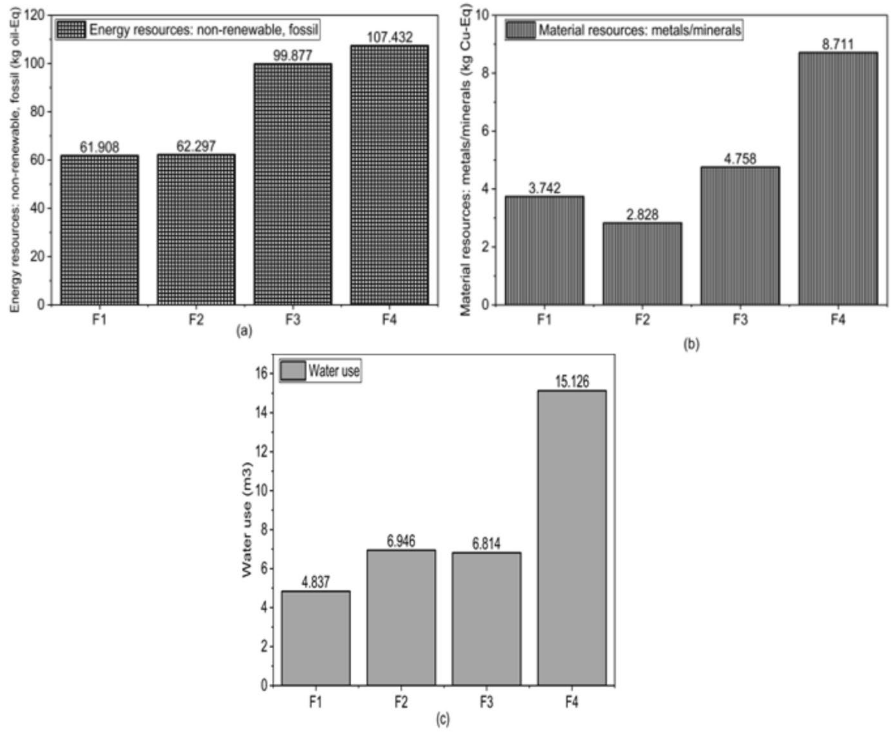

Fig. 4 Mid-point characterization results **a** Non-Renewable Energy Resources, **b** Material Resources, **c** Water use

tailings associated with upstream industrial processes in the manufacture of agricultural machinery, fertilisers, and agrochemicals. F4 exhibited the highest toxicity potentials in both categories, correlating with its heavy reliance on mechanized operations and specific fertilizer formulations, aligning with global concerns over pesticide and heavy metal emissions in rice cultivation [2].

Particulate matter formation, linked to fossil fuel combustion in farm machinery, followed similar patterns, with F4 demonstrating nearly three times higher impact than other farms due to intensive mechanization.

Across all regions, Soil Fertility (fertilizer production and application) and Machinery Fuel (on-farm operations and water pumping) were the dominant contributors to Climate Change and Energy Resources, aligning with results from other emerging economies such as Iran, where LCA studies in high-input systems reflected highest impacts for GWP, acidification, and resource depletion [42, 25]. Furthermore, the production of rice seed consistently drove the majority of the impact for Eutrophication across all farms. Kerala presented a unique profile where specific Ecotoxicity impacts were heavily influenced by micronutrient application, particularly Zinc Sulphate. While fertilizer and fuel are universal hotspots, the most critical targets for regional environmental mitigation strategies must address pesticide use in high-intensity systems like Haryana and the specific micronutrient applications driving ecotoxicity in states like Kerala.

3.1 Comparison of scenarios

Across all impact categories, F4 consistently exhibited the highest environmental burden, identifying it as the least sustainable production system among the studied farms (Table 1). In contrast, F2 achieved the lowest environmental impacts in several categories, particularly those associated with climate change, ecotoxicity, and human toxicity (carcinogenic). This may be related to the better quality of chemical inputs and substantial natural inputs. F3 demonstrated good performance in marine eutrophication, human toxicity (non-carcinogenic), and low ecotoxicity, but moderate impacts in other categories. F1 presented intermediate results, balancing moderate mechanization with traditional practices, resulting in a balance of low and moderate environmental impacts overall. It should be noted that the total relative contributions in the bar charts (F1-F4) may not sum perfectly to 100% and deviations may be attributed to rounding or may indicate that a single process's impact is disproportionately large compared to the sum of all other contributing processes.

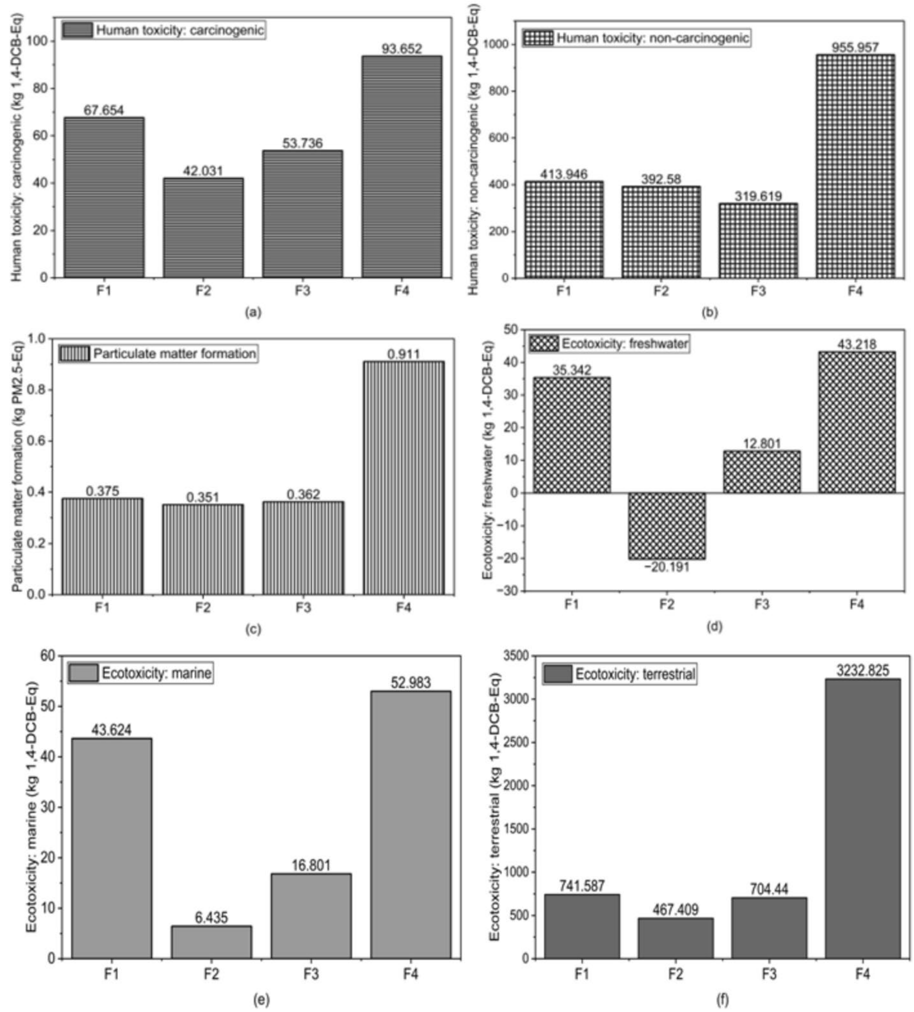

Fig. 5 Mid-point characterization results: **a** Human carcinogenic toxicity, **b** Human non-carcinogenic toxicity, **c** Particulate matter formation, **d** Freshwater ecotoxicity, **e** Marine ecotoxicity, **f** Terrestrial ecotoxicity

Table 1 Comparison of the impacts of the four farms based on their mid-point categories (1 = lowest emissions and 4 = highest emissions)

Impact category	F1	F2	F3	F4
Acidification: terrestrial **(kg SO2-Eq)**	1	3	2	4
Climate change (kg CO2-Eq)	2	1	3	4
Ecotoxicity: freshwater **(kg 1,4-DCB-Eq)**	3	1	2	4
Ecotoxicity: marine **(kg 1,4-DCB-Eq)**	3	1	2	4
Ecotoxicity: terrestrial **(kg 1,4-DCB-Eq)**	3	1	2	4
Energy resources: non-renewable, fossil **(kg oil-Eq)**	1	2	3	4
Eutrophication: freshwater **(kg P-Eq)**	1	2	3	4
Eutrophication: marine **(kg N-Eq)**	2	4	1	3
Human toxicity: carcinogenic **(kg 1,4-DCB-Eq)**	3	1	2	4
Human toxicity: non-carcinogenic **(kg 1,4-DCB-Eq)**	3	2	1	4
Material resources: metals/minerals **(kg Cu-Eq)**	2	1	3	4
Particulate matter formation (kg PM2.5-Eq)	3	1	2	4
Water use (m3)	1	3	2	4

3.2 Uncertainty and variability analysis

The rigorous Monte Carlo simulation, executed over 50,000 iterations, provided a robust, quantitative estimate of result variability for our four distinct farm scenarios. Following the recommendations of Igos et al. [29], the results are presented using the Mean and Median to establish central tendency, alongside the 90% Confidence Interval (CI) to quantify the range of probable outcomes, thereby allowing for precise comparison of statistical significance (Supplementary Table S3). The consistently wide CIs indicate a high degree of model variability and flat distributions, primarily due to the inherent stochasticity of foreground data (e.g., Water Use in F4), reflecting real-world application inconsistencies and diverse distributions and the necessary reliance on generalized background LCI data (e.g., GLO, RoW ecoinvent flows for fertilizer and machinery inputs). Farm F4 consistently exhibited the highest statistical burden across core categories, for example, the Climate Change impact (Mean: 409.06 kg CO_2-Eq; 90% CI: 358.07—467.05) was distinct from the lower-impact Farm F1 (Mean: 243.82 kg CO_2-Eq, 90% CI 213.75—277.04 kg CO_2-Eq), as their respective CIs show no overlap, thereby displaying that the difference in their mean impacts is statistically significant and not merely due to random uncertainty. This probabilistic approach validates the study's conclusions against variations and highlights the critical need for future region-specific LCI data development to constrain estimates in emerging economies.

3.3 Life cycle costing results

Complementing the environmental assessment, the life cycle costing analysis revealed significant differences in the economic costs associated with rice production among the farms. F4 incurred the highest production costs, driven by extensive mechanization, high input use, and equipment rental expenses. F3 demonstrated the lowest overall costs, benefitting from minimal mechanization, low input dependency, and the use of manual labour and scavenged resources (Table 2). F2's high costs stemmed largely from the use of premium imported fertilizers and pesticides, while F1 reported moderate costs, balancing mechanization with local inputs.

Expenditures on natural inputs such as cow dung, jaggery, and tobacco leaves varied widely, with F2 spending the most due to high market prices for cow dung and its transport, whereas F3 minimized costs through local resource collection. Irrigation water was sourced from multiple natural inputs: ponds (F1), a borewell (F2), a government-funded canal (F3), and a well (F4). While the immediate water input cost was uniformly zero (the canal was government-subsidized and provided at no cost), the long-term capital expenditure for borewell/well construction and pump acquisition (dating back decades) could not be verified. Farm F3 uniquely channelled canal water using bunds, requiring no motorized equipment. Operational irrigation costs were therefore restricted to the consumption of diesel (F1, F2) and electricity (F4). Chemical input costs were highest for F2 and F4, reflecting either the use of premium products or high application rates. Labour costs were highest in F3 due to manual harvesting, despite lower daily wages, whereas F2 and F4 managed to reduce labour costs by employing migrant labourers. Fuel and electricity costs were strongly influenced by mechanization levels and transport distances, with F4 and F1 exhibiting the highest expenditures, while F3 recorded the lowest costs due to minimal machinery use and local input sourcing. Equipment rental costs followed a similar pattern, with F4 showing the highest expenses, reflecting its extensive reliance on hired machinery, whereas F1 and F3 minimized rental expenditures through selective mechanization and manual operations.

In summary, the combined environmental and economic analyses reveal distinct trade-offs among the rice production systems studied. While intensive mechanization and high-input strategies, such as those employed by F4, offer potential labour efficiencies, they also incur significant environmental and financial costs. Conversely, farms such as F1 and F3,

Table 2 Life Cycle Costing for the functional unit of each farm from farm to gate in US Dollars ($1 = INR 83.33)

	Cost per 1000 kg Paddy ($1 = INR 83.33)			
	F1	F2	F3	F4
Natural inputs	$25.14 (15.47%)	$29.28 (15.97%)	$9.68 (6.65%)	$7.23 (3.06%)
Chemical inputs	$10.00 (6.15%)	$50.79 (27.7%)	$25.41 (17.45%)	$44.07 (18.66%)
Labour	$62.67 (38.56%)	$39.84 (21.73%)	$72.83 (50.02%)	$67.73 (28.68%)
Fuel	$50.84 (31.28%)	$26.46 (14.43%)	$15.22 (10.46%)	$55.99 (23.71%)
Renting equipment	$13.89 (8.55%)	$36.96 (20.16%)	$22.45 (15.42%)	$61.14 (25.89%)
Total	$162.54 (100%)	$183.34 (100%)	$145.59 (100%)	$236.17 (100%)

which employ more traditional and resource-conserving practices, achieve lower environmental footprints and reduced overall costs, though often at the expense of increased labour requirements. These findings underscore the importance of regionally tailored agricultural strategies that consider both environmental sustainability and economic viability.

4 Discussion

Rice cultivation stands at a complex intersection of environmental impact, food security imperatives, and economic pressures, exacerbated by climate risks and policy landscapes. Effective decarbonisation demands data-driven solutions that assess the full ecological costs of achieving food security beyond GHGEs. Existing global and regional LCA studies highlight the context-specific nature of environmental hotspots and trade-offs in agricultural systems [9, 23, 24, 41]. This study uniquely contributes to the literature by integrating LCA with LCCA, offering a holistic understanding of the environmental and economic implications of agricultural decisions, specifically from the producer's perspective.

4.1 Trade-offs in agricultural systems of emerging economies

While the Green Revolution significantly boosted productivity from 1013 kg/ha to 2882 kg/ha, leading to food self-sufficiency and a reduction in malnutrition to 1 per 1,00,000 by 2021 [21], our findings illustrate how its legacy continues to shape contemporary environment burdens. By 2021, India's agricultural sector was responsible for 755.05 Mt (19%) of the country's 3.95 Gt CO_2-eq of GHGEs, where rice cultivation alone contributed 67.70 Mt CO_2-eq [60, 75]. These trade-offs threaten long-term food security through resource degradation, climate instability, and adverse impacts on biodiversity and human health. Evaluating these trade-offs is complicated by their location-specific nature and varied socio-economic impacts [7]. Yuan et al. [78] showed that achieving high yields and high resource-use efficiency are not conflicting goals. They found systems in South Asia often have large yield gaps but inefficient nitrogen use, but the results in this study showed that high input did not necessarily guarantee higher efficiency in production, leaving room for improvement in resource-use efficiency in India.

Analysis of farming system F2 revealed a large environmental trade-off between localized benefits from reduced synthetic input use and the logistical burdens associated with natural inputs. Although F2 exhibited the lowest impacts for climate change, material resource use (metals/ minerals), human toxicity (carcinogenic), particulate matter formation, and ecotoxicity it ranked third in terrestrial acidification and ranked the poorest for marine eutrophication. This elevated impact was primarily driven by the extensive processes related to cattle rearing, the source of cow manure.

This finding highlights a critical challenge in transitioning to organic agriculture: the environmental burden shifts from the manufacturing of synthetic inputs to the production of bulky natural alternatives, illuminating the challenges of manure management in smallholder contexts particularly where local access is limited [9]. Consequently, promoting natural inputs as a decarbonization strategy for Indian rice cultivation without addressing their environmental and logistical footprint risks creating new environmental externalities and underestimating the system's total environmental cost.

4.1.1 Trade-offs associated with input choices

High mechanization and the application of zinc sulphate monohydrate reportedly enhanced rice yield in F4, but this practice simultaneously elevated terrestrial and freshwater ecotoxicity due to the risk of heavy metal pollution. Zinc emerged as a major hotspot for terrestrial ecotoxicity. As an agricultural runoff, elevated concentrations of zinc are known to increase fish mortality and post a risk to human health by polluting water bodies [13, 59]. Furthermore, its accumulation in soil threatens indigenous microbial communities, which can require costly remediation [40, 70]. Thus, the complex environmental consequences of optimizing for agricultural outcomes requires a holistic assessment of input choices and calculation of the risk–benefit ratio to maintain soil, water, and biodiversity [7, 15]. Conversely, reliance on natural inputs causes yield limitations, as Smith et al. [61] suggests that manure may only provide 52–80% of the requisite nitrogen for farmlands.

4.1.2 Reduced ecotoxicity vs. increased input costs

F2 showed lower impacts on freshwater, marine, and terrestrial ecotoxicity, attributed to its use of higher-quality, synthetic inputs from international brands that reportedly offered greater efficacy at lower dosages, potentially reducing both on-farm and upstream environmental burdens (Table 1). However, this approach incurred a substantial economic trade-off, with F2's chemical input costs ($50.79, 27.7%) being over four times greater than F1's ($10, 6.15%) constituting a major portion of its total expenses (Table 2). In contrast, natural alternatives represented 15% of the budget for both F1 ($25.14) and F2 ($29.28). Kadam et al. [39] reported a similar cost–benefit dilemma that less toxic, biological pesticides in India often necessitate larger quantities or more frequent applications and are hampered by inefficient supply chains and non-standardized pricing, thereby increasing farmers' direct costs. Furthermore, government subsidies for synthetic nitrogen fertilizers encourage overuse, leading to environmental degradation [10, 46]. Although farmers prefer minimizing synthetic inputs, the superior efficacy and reliability of conventional products limit the adoption of sustainable alternatives. However, Vos and Martin [72] suggested that reallocating subsidies toward inputs with lower emission intensities could offer environmental benefits with only minor adverse impacts on economic efficiency.

4.1.3 Decarbonisation vs. labour intensity

F3 achieved the lowest total production cost due to minimal mechanization, gravity-fed irrigation, and scavenging for natural inputs locally. Similarly, F1 maintained lower costs through limited mechanization and chemical inputs. This cost structure, however, was offset by significantly higher labour demands for tasks such as harvesting and weeding. Mechanised farming is often considered energy efficient and environmentally preferable due to higher yields, reducing environmental burden by mass of output, as proven by Khoshnevisan et al. [42]. However, in this study, farms with highest mechanisation (F4) also produced the highest GWP due to the carbon intensive inputs, and highest life cycle costs driven by expensive rentals and fuel, as opposed to F3 with its minimal mechanisation, manual tools and implements, and reliance on human labour due to lower economic strength, thereby providing a socioeconomic benefit for high-labour economies.

Labour constituted the largest share of total input costs for both F1 (38.56%, $62.67) and F3 (50.02%, $72.83). This trade-off between capital and labour could be a socioeconomic benefit in emerging economies like India, creating rural employment supporting livelihoods threatened by climate change [68]. The primary economic advantage of employing a migrant workforce by F1, F2, and F4 stem from their ability to meet the high, seasonal labour demands, thereby mitigating crop loss and ensuring a stable food supply. This flexible, cost-effective labour pool directly contributes to lower production costs for farmers. Bui and Le [8] further found that labour intensive farming improved farm performance when measured in terms of income for the farmer and workers. Expanding the job market highly supports rural communities in emerging economies like India, where the rural Labour Force Participation Rate remained 54.8%, and climate change could exacerbate unemployment [49]. These findings present a direct trade-off between capital investment in mechanization for decarbonization and labour-intensive practices. This contrasts with conventional support for mechanization but is contextualized by the circumstances of smallholder farmers in India, who face prohibitive rental costs for machinery on fragmented and difficult terrain. In this context, supporting local labour appears to be more beneficial.

In regions like India, that sees an exodus of labourers' movement across states (the 'agriculture-migration nexus') during plantation and harvest seasons, agricultural mechanisation threatens jobs and rural livelihoods [54]. However, this also raises concerns about worker exploitation, substandard working conditions, and limited bargaining power. The resulting lower food prices enjoyed by consumers are effectively subsidized by compromised wages and substandard living conditions of the workforce. This structural inequity poses a direct challenge to achieving equitable wages and upholding human dignity, contradicting the principles of Sustainable Development Goal 8 (Decent Work), where food security is prioritised over labour equity. A critical contributing factor is the persistent issue of regulatory gaps. Many jurisdictions lack effective mechanisms for monitoring and enforcing labour standards in seasonal farm work, enabling persistent violations. While cheap migrant labour can depress local agricultural wages, improved compensation could bankrupt small farms under unstable climatic conditions. The challenge for policymakers is balancing the need for low food costs with the necessity of ensuring equitable wages and human dignity.

4.1.4 Financial trade-offs in agricultural practices

Mechanization in F4 led to higher overall production costs and environmental impacts due to increased spending on fuel, electricity, and equipment rental. In contrast, other farms demonstrated distinct economic trade-offs. F3's reduced mechanization was counterbalanced by high labour costs, which constituted half of its budget ($72.83, 50.02%) despite lower regional labour rates and family assistance. Economic viability remains a primary constraint for sustainable agriculture, as farmers facing rising operational costs. F1 exhibited the lowest chemical input costs ($10, 6.15%) but comparatively high natural input cost ($25.14, 15.47%), whereas F2's use of premium synthetic pesticides incurred the highest chemical expenditure ($50.79, 27.7%). F3 minimized costs ($9.68, 6.65%) by scavenging for local natural inputs, highlighting the financial benefits of localized resource acquisition.

Reported returns from rice production (USD 360 to USD 720 per cropping cycle) often fall below the average monthly liveable wage of USD 185, primarily due to high input costs, market price fluctuations, climate risks, and transport expenses [45]. Furthermore, price regulations aimed at maintaining food affordability often suppress farmgate prices, resulting in some farmers withholding produce for household consumption or barter (F1), thereby reducing contributions to national food security.

4.2 Policy implications and regional adaptation strategies

Global evidence confirms that agricultural sustainability transitions depend on robust policy frameworks and sustained public investments as short-term financial incentives alone are often insufficient [50]. Governments in India and ASEAN countries have developed strategies towards promoting and research in sustainable agriculture, but face severe bottlenecks in practice owing to changing operational costs, price instability, unpredictable earnings, climate-driven disasters, and lack of knowledge and training among producers. Studies recommend crop insurance and targeted compensation for yield losses during the transition to sustainable agricultural practices [35, 65]. Achieving the dual goal of lowering agricultural emissions while ensuring farmer livelihoods and food security requires a complex range of interventions, particularly in emerging economies with diverse and difficult-to-monitor agro-ecological contexts [72].

The outcomes in this study suggest the need for agro-ecological zonation and region-specific strategies such as targeted investments in rural composting, methane management, water reforms, and labour accessibility and easier access to climate finance and carbon markets for rice farmers adopting methane-mitigation and low-emission practices [72].

4.2.1 Balanced input subsidy reforms

F3 exclusively used highly-subsidized urea, omitting costlier inputs like diammonium phosphate. This cost-driven choice likely resulted in nitrogen-skewed NPK ratios in the soil [18, 46]. Gautam et al (2022) [18] reported that the current global government subsidies often incentivise unsustainable agriculture, contributing to 22% GHGE, and threatening to raise them by 58%. These powerful financial incentives highlight the need to rebalance agricultural subsidies, shifting support away from conventional synthetic inputs towards safer alternatives, such as low-ecotoxicity premium products and biofertilizers. However, policy evaluation by Springmann and Freund [62] identified complex trade-offs where policies that could promote sustainable agriculture raised food costs, while those benefiting environmental and public health may not adequately support producers, suggesting a need to shift subsidies from a production-centric model to one focused on health and environmental outcomes, accounting for future climate-modified agricultural systems. Directing subsidies toward foods with positive health and environmental profiles could shift consumer demand. Investing in sustainable innovations like climate-resilient technologies may offer a comprehensive solution for environmental, social, and economic goals, offering a 'win–win-win' scenario for the people, the planet, and prosperity [79].

4.2.2 Supply chain and logistical innovations

The logistical challenges of transporting bulky organic inputs, such as manure for F1 and F2, shift environmental burdens from manufacturing to transportation. This highlights the need for investment in decentralized, locally managed composting facilities to reduce transport-related GHGEs and stimulate local economies. A robust manure-management policy that establishes local standards and networks for collection, processing, and safe storage could minimize both transport and manure-related emissions, the latter being a significant source of agricultural GHGEs [17, 75].

However, Kumar et al. [43] cautioned against abrupt, large-scale shifts toward organic production without ensuring reliable substitutes for synthetic inputs, citing risks of significant yield reductions. Furthermore, premium pricing models for organic produce, while intended to enhance farmer profitability, could create market access barriers for lower-income consumers, raising significant equity concerns and complicating the scalability of sustainable agriculture in emerging economies.

4.3 Decarbonisation, food production, and SDGs

Achieving Sustainable Development Goals (SDGs) in agriculture requires evidence-based policies informed by holistic data obtained from LCA and LCCA. This study identified key intervention hotspots, including GHGE from the transport of natural inputs (F1, F2), mechanisation and the use of specific agricultural inputs (F4).

Addressing these hotspots through decentralized facilities could reduce emissions and create local employment, supporting SDG 13 (Climate Action) and SDG 8 (Decent Work and Economic Growth). While labour-intensive practices increase wage costs, they boost rural employment, a crucial factor for SDG 2 (Zero Hunger). Wilbois and Schmidt [74] further suggest reframing the debate on farming methods away from a yield-oriented approach to one that focuses on a wider range of values that include biodiversity and conservation.

Transport costs and low market prices often limit farmers to subsistence levels, impeding broader contributions to food security. The analysis of inherent trade-offs—synthetic vs. organic inputs, mechanization vs. labour, and ecotoxicity vs. cost—is central to the sustainable management of resources and chemicals (SDG 12.2, 12.4). Given that rice cultivation emits significantly more GHGE than other cereals, decarbonization and methane mitigation are critical priorities for SDG 13 [55]. Addressing agricultural runoff and aquatic toxicity directly supports the protection of water bodies (SDG 6) and terrestrial ecosystems (SDG 15).

Consumer behaviour analyses in Asia indicate a promising market for sustainably produced food (SDG 12), driven by health and safety concerns followed by concern for the environment, but cost barriers and limited accessibility, hinder this target [4, 51, 52] in press). Advancing these goals requires interdisciplinary collaboration to reform agricultural systems, prioritizing environmental outcomes alongside production and cost objectives [23, 24]. Ultimately, effective policy hinges on high-quality, regionally relevant, and inclusive life-cycle data to identify unique hotspots and guide evidence-based interventions that address both environmental sustainability and socioeconomic equity.

4.4 Limitations and recommendations

This research primarily relied on international life cycle inventory datasets, particularly the ecoinvent database, due to the limited availability of region-specific data in emerging economies. Select parameters—such as rice seed production, water pump operation, and electricity generation from coal—were based on localized data sources. This reliance on non-India-specific background data is a limitation of the study, and the quantified environmental impacts often reflect a global average of production technology rather than specific Indian regional production processes. Consequently, impact estimates for categories including fertilizer production, pesticide manufacturing, and agricultural machinery use may be subject to uncertainty. Future efforts for environmental improvement should focus enhanced traceability and substitution of high-impact imported inputs. To support this, subsequent research must prioritise the development of region-specific LCI data for key agricultural inputs such as fertilizers, pesticides, energy systems, and machinery to improve the accuracy and relevance of national-level sustainability assessments.

5 Conclusion

This study, through its unique integration of Life Cycle Assessment and Life Cycle Cost Analysis, provides a holistic and nuanced understanding of the environmental and economic implications within emerging agricultural economies, specifically in rice cultivation. This research demonstrates that high-input, mechanized rice systems, while often efficient, result in elevated environmental impacts, particularly in terms of greenhouse gas emissions, acidification, and ecotoxicity. Key drivers include fossil fuel dependency, synthetic fertilizer use, and transport logistics. Our findings reveal critical trade-offs that demand urgent, evidence-based policy action to achieve Sustainable Development Goals.

We identified several novel insights. Firstly, while the shift to natural inputs offers environmental benefits, it introduces a significant transport and cost burden, transforming the environmental hotspot from input manufacturing to logistics, a previously overlooked challenge in smallholder contexts. Secondly, high mechanization and the application of zinc sulphate, while enhancing yields, lead to a concerning ecotoxicity hotspot in aquatic and terrestrial environments, posing risks to biodiversity and human health. Thirdly, the perceived efficacy of often-imported, higher-quality synthetic inputs, despite their lower ecotoxicity, translates into significantly increased costs for farmers, disincentivizing the adoption of safer, domestically available alternatives due to poor supply chains and non-standardized pricing. Finally, our study highlights a direct trade-off between decarbonization and labour intensity, where reduced mechanization, while lowering environmental impact and production costs, increases labour demands. This, however, can positively contribute to rural employment and livelihoods in highly populated emerging economies, offering a unique perspective that contrasts with common mechanization narratives. The findings of this research provide evidence that sustainable rice production in emerging economies requires a multidimensional approach that integrates environmental, economic, and social considerations.

Rice cultivation accounts for 1.5% of global greenhouse gas emissions, yet climate mitigation is hampered by a lack of region-specific data LCA data paramount for identifying specific hotspots and guiding targeted interventions towards a circular economy of resource transformation and replenishment. Achieving sustainability goals requires policies that rebalance agricultural subsidies, shifting focus from synthetic inputs like urea towards cleaner, low-ecotoxic alternatives, including biofertilizers and locally sourced natural inputs. Simultaneously, supporting labour-intensive farming systems could bolster SDG 8 (Decent Work and Economic Growth) by generating crucial rural employment. Ultimately, achieving SDG 2 (Zero Hunger) necessitates a balanced and economically viable approach that sustains yields while transitioning to low-emission practices. The complexities revealed underscore that progress towards these SDGs remains uneven, necessitating context-specific, inclusive, and multi-stakeholder agricultural reforms guided by high-quality, regionally relevant, evidence-based data. Future research must integrate Life Cycle Assessment, Life Cycle Cost Analysis and advanced decision-support tools, such as artificial intelligence models trained on localized data, to optimize interventions at the farm level. Ultimately, sustainable rice production in emerging economies must extend beyond yield maximization to encompass the entire agricultural value chain, ensuring long-term resilience and alignment with global sustainability targets.

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Declaration of generative AI and AI-assisted technologies in the writing process During the preparation of this manuscript, the authors utilized a large language model, Gemini, strictly for improving the quality of the language, clarity, and readability of the text. Gemini's specific contributions were limited to refining the syntax, grammar, and sentence structure of sections drafted entirely by the authors, and rephrasing texts to meet the journal's requirements for clarity and length. The authors have reviewed, edited, and verified all content generated by Gemini to ensure accuracy and fidelity to the original scientific intent. AI was not involved in designing the methodology, data processing, or interpretation. The core scientific content, methodology, findings, analyses, and conclusions remain the sole intellectual property and responsibility of the authors.

Author contributions 1. Rachael Alphonso: Conceptualisation, Data curation, Formal analysis, Funding acquisition, Investigation, Methodology, Validation, Visualisation, Writing–Original Draft. 2. Thirumani Devi Arumugam: Project administration, Supervision. 3. Venkata Ravi Sankar Cheela: Conceptualisation, Data curation, Formal analysis, Methodology, Validation, Visualisation, Project administration, Supervision, Writing–review and editing. All authors have read and approved the final manuscript.

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Data availability The data that support the findings of this study are currently under an embargo as they form part of an ongoing doctoral dissertation. The complete raw dataset will be deposited in a national public repository and made fully available upon the successful defence and submission of the final thesis. The data that support the findings will be available from the corresponding author upon reasonable request.

Declarations

Ethics approval and consent to participate This study was performed in accordance with the ethical standards as laid down in the 1964 Declaration of Helsinki and its later amendments. The questionnaire and methodology for this study were approved by the Institutional Human Ethics Committee of Avinashilingam Institute for Home Science and Higher Education for Women, Coimbatore, India. The approval number is AUW/IHEC/FSN-21-22/XPD-40. The researchers obtained informed consent from all participants included in the study for their participation in this study.

Consent for publication The researchers obtained informed consent from all participants included in the study for their consent to publish the data collected from the interviews. No identifying data has been included in this paper.

Competing interests The authors declare no competing interests.

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